Linear Gaussian Bayesian Network for Mortgage Approval Decision

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1 Motivation

This project was created during Harvard's CS50's Introduction to Artificial Intelligence with Python course [2], as I wanted to explore Bayesian networks in more depth since they were only covered briefly in the curriculum.

2 Introduction

This project simulates an internal mathematical model for Bayesianhill Bank that trains a Linear Gaussian Bayesian Network (LGBN) with 27 nodes using the pgmpy [1] library on CSV data from clients who have already repaid or defaulted on their mortgages. Based on this historical data, the model calculates the probability that a new mortgage applicant will successfully repay their loan when a banker inputs their information into the system, including loan amount and repayment period to assess risk based on borrowing parameters.

3 Results

The model's performance was evaluated using a rigorous two-stage methodology. First, an independent test dataset was generated using the "economic downturn" scenario with parameters differing from the training data: 15% lower average salaries (29,750 CZK vs. 35,000 CZK) and 30% higher interest rates (5.85% vs. 4.5%). This stress-testing approach ensures the model's robustness under adverse economic conditions. Ground truth labels were established using comprehensive financial criteria including payment-to-income ratios, debt-to-income ratios, credit history, employment stability, and housing status, rather than relying on model-generated outputs.

Second, 5-fold stratified cross-validation was performed on the training data, where each fold involved training a complete Gaussian Bayesian Network from scratch and evaluating on the validation subset. This process validates the model's consistency and generalization capability across different data partitions.

The evaluation generates several key outputs: a confusion matrix visualization showing true vs. predicted classifications, an ROC curve demonstrating the model's discriminative ability across different probability thresholds, and comprehensive metrics including accuracy, precision, recall, F1-score, and AUC. All results are automatically saved to the evaluation_results directory for detailed analysis.

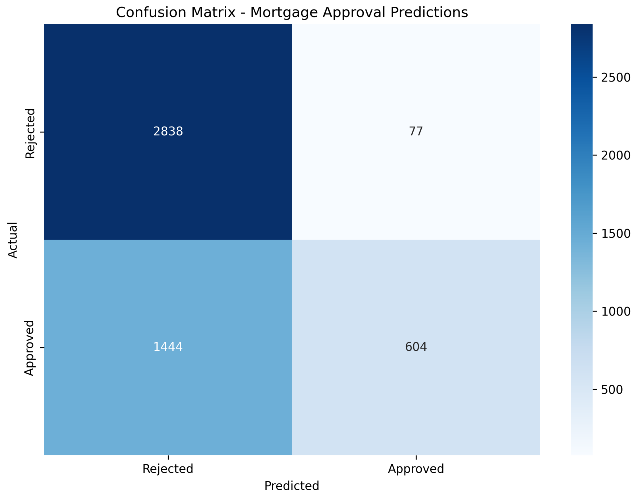

Figure 1: Confusion Matrix showing model predictions vs. actual loan approval decisions

Performance metrics from the most recent evaluation on 4,963 test samples revealed: Accuracy of 69.4%, Precision of 88.7%, Recall of 29.5%, F1-Score of 44.3%, and ROC AUC of 67.5%. Cross-validation yielded consistent results with a mean accuracy of 68.4% ± 0.5% across all folds, confirming the model's stability. The high precision indicates that when the model approves a loan, it is correct 88.7% of the time, though the low recall shows it conservatively approves only 29.5% of actually qualified applicants, reflecting typical risk-averse behavior in financial institutions.

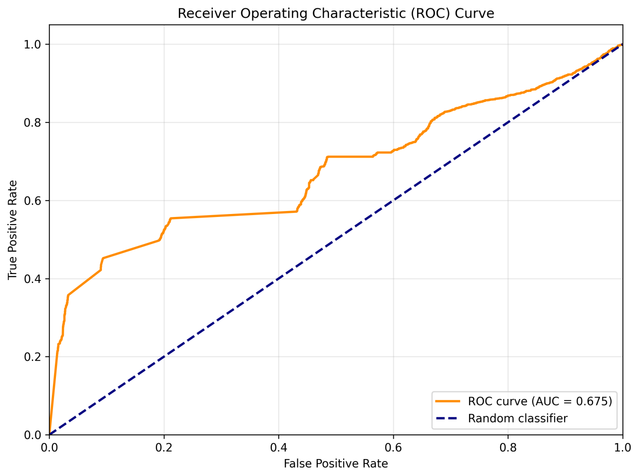

Figure 2: ROC Curve demonstrating the model's discriminative performance (AUC = 0.675)

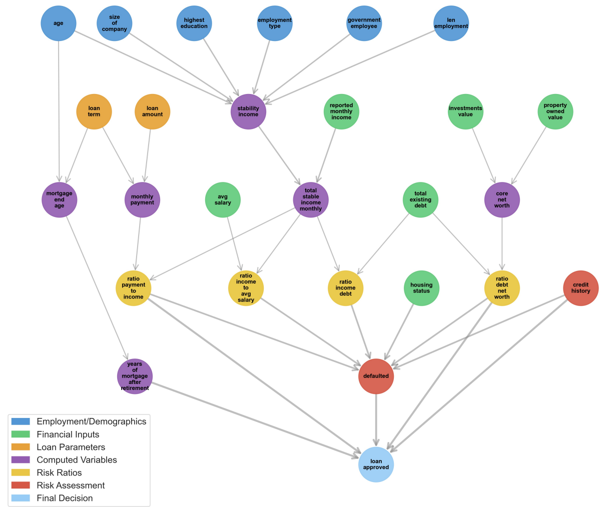

Figure 3: Diagram of the 27-node LGBN

4 Training

The model is trained on a CSV file that is synthetically yet realistically generated, though the program also offers training on any other arbitrary CSV file within the folder. The goal of this project was to implement and familiarize myself with Bayesian Networks (initially I experimented with Discrete Bayesian Networks using the Pomegranate library). It is important to note that the model is trained on synthetic data and evaluates mortgage approval probability based on this training, which should be taken into consideration. Nevertheless, the model is highly realistic, and I successfully fine-tuned it to demonstrate the practical power of Bayesian Networks in a compelling and realistic manner.

5 Input Data

The input data, provided by an employee of Bayesianhill Bank, contain basic information about the mortgage applicant. Specifically: age, highest level of education, number of employees in the applicant's company, years of employment with the current employer, type of employment contract, whether the applicant is a public sector employee, net monthly income, total debt, value of real estate and investments, credit score (excellent, fair, bad), and housing status (rent, own, mortgage, homeless, etc.). In addition, the program requires basic information about the average salary, the amount of the requested loan, and the intended repayment period (from 1 to 35 years). All other nodes are computed automatically.

In the diagram, the light-blue node labeled *Loan Approved* represents the final node, which returns the probability that the applicant will successfully repay the mortgage in full.

6 Data Generation

A synthetic dataset was generated with the aim of closely approximating realistic conditions. This dataset was subsequently used for training the LGBN model. The generation process employed the *random* and *time* libraries to enhance efficiency, with stochastic elements playing a central role. The procedure began with the random assignment of applicants' ages, followed by the derivation of their highest level of education as a function of both age and randomness. The same stepwise approach was applied to the remaining features, ensuring interdependencies were preserved where appropriate.

7 Representation of Program Result

Let us now illustrate the program execution and its evaluation. The interest rate is set at 4% and remains constant throughout the mortgage period. The average monthly salary is set to 35,000 CZK, and retirement age is assumed to be 65 years. The LGBN model was trained on 1,000,000 rows of data.

The mortgage approval probability is 100.0% (corresponding to Table 1), which is not surprising, as the monthly payment is small relative to the applicant's income.

When the loan term is increased from 30 to 35 years (see Table 2), the model, trained on the data, recognizes that a mortgage extending beyond the retirement age (in our case, only 5 years past retirement) is riskier. Consequently, the mortgage approval probability is evaluated at 50.4%.

Table 1: Mortgage Applicant A

Attribute	Value
Age	35 years
Government Employee	No
Education	Master
Student Status	No
Employment Type	Permanent
Years in Company	6 years
Company Size	240 employees
Monthly Income	60,000 CZK
Existing Debt	0 CZK
Investments	120,900 CZK
Property Value	0 CZK
Housing Status	Rent
Credit Score	Excellent
Loan Amount	2,000,000 CZK
Loan Term	30 years
Interest Rate	4.0% p.a.
Monthly Payment	9,548 CZK

Table 2: Mortgage Applicant A

Attribute	Value
Loan Amount	2,000,000 CZK
Loan Term	35 years
Interest Rate	4.0% p.a.
Monthly Payment	8,855 CZK

Table 3: Mortgage Applicant A

Attribute	Value
Loan Amount	12,000,000 CZK
Loan Term	30 years
Interest Rate	4.0% p.a.
Monthly Payment	57,290 CZK

Alternatively, if the loan amount is increased (see Table 3) to 12 million Czech Crowns and the repayment term is shortened to 30 years so that the model does not penalize repayment beyond the usual retirement age, the mortgage approval probability is evaluated at 2.4%.

Let us consider a completely new example in Table 4, demonstrating the patterns that the LGBN has learned. This time, with extreme debt, the applicant is otherwise an ideal candidate for a mortgage but is heavily indebted, which is inconsistent with their income and assets, as they have no investments or property. The model learned this pattern from the CSV data and evaluated the applicant with a mortgage approval probability of 0.0%.

Table 4: Mortgage Applicant B

Attribute	Value
Age	35 years
Government Employee	Yes
Education	Master
Student Status	No
Employment Type	Permanent
Years in Company	9 years
Company Size	390 employees
Monthly Income	45,000 CZK
Existing Debt	9,000,000,000 CZK
Investments	0 CZK
Property Value	0 CZK
Housing Status	Rent
Credit Score	Excellent
Loan Amount	100,000, CZK
Loan Term	30 years
Interest Rate	4.0% p.a.
Monthly Payment	443 CZK

8 Methodology

During implementation, I faced several challenges, primarily because LGBN assumes linear relationships, which may not hold in reality. Financial client data often exhibits "heavy tails" and outliers—extreme values that degrade LGBN stability.

I encountered stability issues and therefore attempted to train the model without extreme values relative to the rest of the dataset, which subsequently significantly complicated the model's ability to produce realistic results. It was very difficult to correctly calibrate the data generation to produce appropriate data containing realistic LGBN-capturable patterns across various threshold values—this task proved more challenging than I initially anticipated.

I therefore chose a *hybrid approach*, where the LGBN model returns a probability which I subsequently multiply by a constant based on the applicant's credit history (excellent, fair, bad) and also multiply by a constant less than 1 when the mortgage applicant would exceed the retirement age threshold in the future. The LGBN provides a structured baseline while hand-crafted rules capture domain expertise.

9 Conclusion

I worked on this project in the summer of 2025, and it gave me a lot of valuable experience, including the previously mentioned course [2]. Looking back, I somewhat regret choosing to implement an

LGBN model for the task of predicting mortgage approvals, as I believe it was not the most suitable representation for this type of problem. Regression models, XGBoost, or even Neural Networks would likely have been a better fit. Nevertheless, I do not regret the choice overall, since I see this project as another step in exploring probabilistic models, and I firmly believe that the LGBN model deserved a place in my journey.

References

[1] Ankur Ankan, Abinash Panda, and contributors. pgmpy: Probabilistic graphical models using python. online, 2015. [cit. 2025–08–24] `https://pgmpy.org/`.

[2] Harvard University. Cs50's introduction to artificial intelligence with python. online, 2025. [cit. 2025–08–24] `https://cs50.harvard.edu/ai/`.